

VIKING AGCM, AB, Canada

WMO: 718880

Lat: 53.182N

Lon: 111.732W

Elev: 2333

StdP: 13.50

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-22.9	-18.4	-29.0	1.2	-22.6	-24.1	1.6	-18.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	81.8	62.3	79.0	61.3	76.0	60.2	66.5	76.9	64.4	74.9	62.4	72.1	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.6	92.5	72.7	60.3	85.2	69.5	58.4	79.6	67.3	32.5	76.7	30.7	74.7	29.3	72.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-29.4	87.5	3.4	2.9	-31.9	89.6	-33.9	91.3	-35.8	92.9	-38.3	95.1		
(5)			-29.7	71.2	3.3	2.4	-32.1	72.9	-34.0	74.3	-35.8	75.7	-38.2	77.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.8	12.2	10.2	22.6	37.3	51.5	58.3	62.0	61.0	51.9	39.1	22.8	10.5		
	DBStd	22.09	14.81	14.15	13.34	9.51	8.19	5.66	4.69	5.83	8.27	9.03	12.81	13.62		
	HDD50	6087	1171	1116	850	392	81	9	0	2	74	354	815	1223		
	HDD65	10371	1636	1536	1315	831	422	210	118	148	398	804	1265	1688		
	CDD50	1255	0	0	0	10	127	259	371	343	130	15	0	0		
	CDD65	65	0	0	0	0	3	10	24	24	4	0	0	0		
	CDH74	1062	0	0	0	4	122	134	295	374	130	3	0	0		
	CDH80	173	0	0	0	0	18	15	49	69	22	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	16.1	0.9	0.5	0.6	1.2	1.6	2.8	2.9	2.3	1.3	0.7	0.7	0.7		
	PrecMax	23.1	2.8	0.9	1.3	3.7	2.6	4.9	5.9	4.9	3.1	1.3	3.3	1.3		
	PrecMin	8.5	0.2	0.0	0.2	0.2	0.2	0.6	1.1	0.4	0.1	0.0	0.1	0.2		
	PrecStd	3.3	0.6	0.3	0.3	0.8	0.6	1.1	1.3	1.0	0.9	0.4	0.7	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	41.5	53.4	71.5	82.6	81.7	85.3	85.8	83.2	71.7	55.2	41.1		
		MCWB	36.2	36.3	43.0	49.4	55.5	60.9	67.6	63.2	60.3	54.2	44.5	35.7		
	2%	DB	39.1	37.2	46.5	64.4	77.6	77.8	80.6	81.1	77.8	62.6	46.8	37.1		
		MCWB	34.6	33.4	39.5	47.2	54.2	59.5	66.0	62.9	57.3	49.3	39.7	33.1		
	5%	DB	36.6	34.2	41.6	58.0	72.7	74.1	77.2	78.5	73.3	57.5	42.0	33.4		
		MCWB	33.1	31.3	36.6	44.1	52.7	58.1	63.8	62.7	56.3	46.5	36.8	29.9		
	10%	DB	33.0	31.0	38.1	52.7	68.2	70.9	74.0	74.7	67.5	52.9	38.1	29.2		
		MCWB	30.1	28.7	33.9	41.5	50.2	56.6	62.3	61.5	53.9	44.1	34.1	26.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.8	36.3	44.3	51.2	57.6	65.3	71.2	69.4	62.4	55.0	45.2	36.2		
		MCDB	41.2	40.1	51.8	70.1	76.4	75.8	80.7	78.3	76.4	68.8	53.2	40.5		
	2%	WB	35.1	33.9	39.7	47.5	55.4	62.5	67.5	66.2	59.2	51.1	40.3	33.2		
		MCDB	38.7	36.6	45.4	62.4	71.9	73.5	78.0	77.2	74.1	59.6	46.2	36.9		
	5%	WB	33.0	31.4	36.7	44.8	53.7	60.3	65.2	63.9	56.7	47.7	37.1	29.9		
		MCDB	36.3	34.0	41.5	56.9	69.6	70.1	75.1	74.6	70.2	55.9	41.2	33.2		
	10%	WB	30.1	28.6	34.2	42.2	51.9	58.3	63.2	62.0	54.7	44.8	34.5	26.6		
		MCDB	32.9	31.1	37.8	51.4	65.9	67.7	72.1	72.5	66.7	52.4	37.9	29.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	16.8	16.2	18.8	22.6	20.0	20.2	22.1	22.2	18.6	15.1	14.5		
		MCDBR	17.1	18.7	19.0	28.7	29.3	25.4	25.8	28.3	31.7	26.7	19.6	18.7		
		MCWBR	14.0	15.7	13.2	14.8	11.9	11.1	13.4	13.0	14.3	14.8	13.3	15.4		
	5% WB	MCDBR	17.2	18.1	18.1	26.5	25.9	21.1	23.7	24.7	28.4	24.7	18.0	18.5		
		MCWBR	14.4	15.3	12.8	14.3	11.2	10.7	13.3	12.6	13.7	14.3	12.8	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.279	0.301	0.328	0.347	0.364	0.360	0.372	0.311	0.276	0.251	0.229		
	taud		2.308	2.288	2.346	2.369	2.363	2.352	2.383	2.327	2.473	2.543	2.449	2.350		
	Ebn at Noon		237	264	282	287	286	281	280	269	277	268	241	229		
	Edh at Noon		18	26	30	34	36	37	35	35	27	20	16	14		
(44) All-Sky Solar Radiation	RadAvg		322	657	1131	1492	1840	1896	1969	1608	1128	658	322	239		
	RadStd		33	49	88	136	127	132	103	120	121	88	35	23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page